MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE NAME	

CIVICS GROUP	1	7	S				INDEX NUMBER	

H2 BIOLOGY 9744

Practical Exam

28 August 2018
2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams and graphs.
Do not use staples, paper clips, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For examiner's Use	
1	/ 21
2	/ 18
3	/ 16
Total	**/ 55**

This document consists of **22** printed pages.

[Turn over]

Answer **all** questions

QUESTION 1

Many plant cells contain a water soluble molecule called ascorbic acid, which has many functions to help the plant survive.

You are provided with an extract from plant cells, **P**, which contains ascorbic acid.

Visking tubing, **V**, is partially permeable, similar to a cell surface membrane, so that some biological molecules will diffuse through the wall of the tubing.

You are required to investigate the diffusion of ascorbic acid from **P** into the water surrounding the Visking tubing over a period of 15 minutes.

Fig. 1.1 shows the apparatus before the water was added.

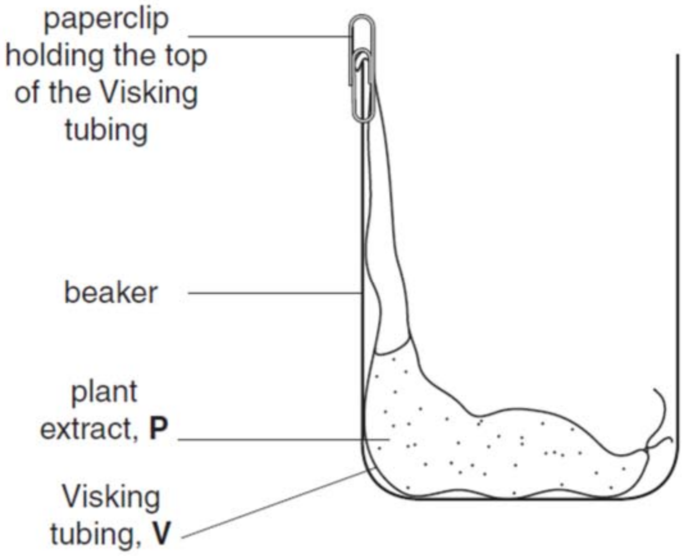

Fig. 1.1

(a) (i) Water is added to the beaker in Fig. 1.1.
Describe the expected trend in the concentration of ascorbic acid in the water over a period of 15 minutes.

-- [1]

You are provided with:

labelled	contents	hazard	volume / cm^3
A	sample of water removed after 15 minutes	irritant	15
P	plant extract containing ascorbic acid	irritant	15
W	distilled water	none	100
I	iodine in potassium iodide solution	irritant	20
S	starch	none	20

labelled	contents	hazard	details	quantity
V	Visking tubing	none	18 cm length in a beaker containing water	1

You must now read up to the end of step 23 before proceeding.

To compare the concentration of ascorbic acid in the samples you are required to find the **volume** of iodine solution, **I**, added to each sample until the end-point is reached.

The drops of **I** will be added one at a time using a small syringe.

To practise releasing drops from a small syringe:

1. Fill the syringe with 2 cm^3 of **I**.

2. Hold the syringe over an **empty** test-tube and push the plunger **gently** to release one drop at a time, as shown in Fig. 1.2.

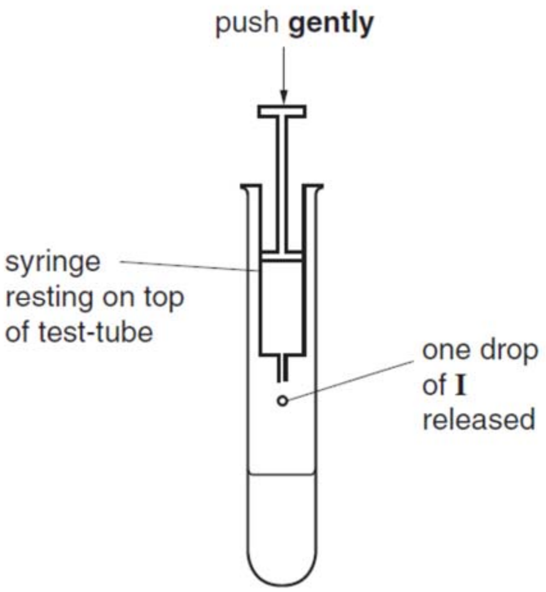

Fig. 1.2

To compare the concentration of ascorbic acid in the samples you will need to add drops of **I** until a blue colour appears. When this blue colour lasts for more than 10 seconds, this indicates the end-point and the **volume of I** that has been added should be recorded.

The apparatus in Fig. 1.1 was set-up and water was added and left for 15 minutes.
Sample **A** was removed from the water in the beaker.

You are required to find the volume of **I** needed to reach the end-point for sample **A**.

Proceed as follows:

3. Put 1 cm^3 of **S** into a test-tube.
4. Put 3 cm^3 of the sample (e.g. **A**) into the same test-tube.
5. Shake the test-tube gently to mix the contents.
6. Fill the syringe, labelled **I**, with 2 cm^3 of **I**.
7. Wipe off any drops of **I** from the outside of the syringe with a paper towel.
8. Add **one** drop of **I** to the mixture in the test-tube as shown in Fig. 1.2.
9. Mix gently and if there is no colour change, add another drop.
10. Continue adding drops, one at a time, until the blue colour appears.
    Wait 10 seconds to see if the end-point has been reached. If the blue colour disappears then add another drop.
11. Repeat step 10 until the mixture stays blue for at least 10 seconds.

(ii) Record the **volume of I** needed to reach the end-point.

volume ………………………… [1]

You are required to:

- set up a Visking tubing containing **P** as shown in Fig. 1.3
- decide the level of water to put into the beaker
- remove samples of the water surrounding the Visking tubing at **5 minute intervals** for 15 minutes
- compare the ascorbic acid concentrations in the samples.

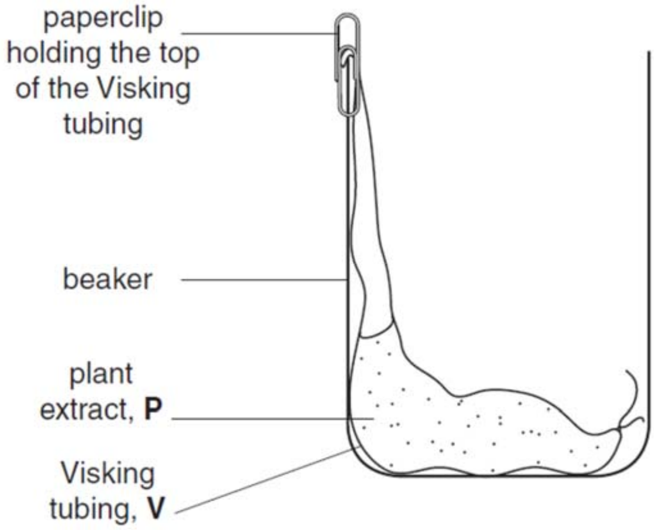

Fig. 1.3

Samples of water surrounding the Visking tubing will be removed for testing, so you need to take this into account when you decide the level of water to put into the beaker.

(iii) Draw **on Fig. 1.3** the level of the water

- before you removed any samples (label 'before'),
- after the total volume of water needed for all the tests has been removed (label 'after').

[2]

(iv) In order to compare the ascorbic acid concentrations, state **one** variable which you will need to standardise when finding the volume of **I** added to each sample.

Describe how you will standardise this variable.

variable: ....

description: ....

....

[1]

Proceed as follows:

12. Put **S**, as in step 3, into the four test-tubes you will require in order to test the samples of water.

13. Tie a knot in the Visking tubing as close as possible to one end so that it seals the end.

14. To open the other end, wet the Visking tubing and rub the tubing gently between your fingers.

15. Put 6 cm^3 of **P** into the open end of the Visking tubing.

16. Rinse the outside of the Visking tubing by dipping it into the water in the container labelled **V**.

17. Put the Visking tubing into an empty beaker as shown in Fig. 1.3.

18. Make sure the open end of the Visking tubing is held in place by a paperclip.

You will start timing as soon as you add **W** (**steps 19 and 20**).

You should read steps 19 to 23 before proceeding.

19. Put **W** into the beaker to the level you decided in **(iii)**.

20. Immediately start timing and remove the first sample of water (as in **step 4**) and put into a prepared test-tube (as in **step 12**).

21. Test the sample as in **steps 5 to 11**.

22. After 5 minutes, gently mix the water surrounding the Visking tubing and then remove the next sample, put it into a different (prepared) test-tube and repeat **steps 5 to 11**.

23. Repeat **step 22** for two more samples.

Record your results in **(a)(v)** on page 7.

(v) Record your results in the space below.

[5]

(vi) Describe how the results support your expected trend as stated in **(a)(i)**.

-- [1]

This investigation provides results to compare the concentration of ascorbic acid in the samples.

(vii) If you had been provided with 1.0% ascorbic acid solution, suggest how you would modify this investigation to find the **percentage concentration** of ascorbic acid in the water after 15 minutes.

....

....

....

....

....

....

....
[2]

(b) Iodine solution (iodine in potassium iodide solution) turns blue-black when starch is present in plant tissues.

However, as ascorbic acid is also found in plant tissues, some scientists investigated the effect of testing for starch with iodine solution when there was ascorbic acid present.

The concentration of ascorbic acid was 0.0001 mol dm^{-3} and the concentration of starch solution was standardised.

The percentage of starch which reacted with the iodine solution was measured.

The results are shown in Table 1.1.

Table 1.1

volume of iodine solution / cm^3	**percentage of starch which reacted with iodine solution / %**
0.0	0.0
0.5	2.0
1.5	5.0
2.0	36.0
2.5	68.0

(i) Plot a graph of the data shown in Table 1.1.

*You will need to consider the answer to **(b)(ii)** before you plot your graph.*

[4]

(ii) Estimate the volume of iodine solution needed for 100% of the starch to be reacted. Show on your graph how you obtained the volume of iodine solution.

volume of iodine solution .... cm^3 [1]

(iii) Explain how the presence of ascorbic acid may affect the use of iodine solution as a test for the presence of starch in different plant tissues.

[2]

(iv) A plant tissue contains starch and 0.0001 mol dm^{-3} ascorbic acid.
Suggest how you would make sure that the iodine test showed the presence of all the starch (100%).

----
[1]

[Total: 21]

QUESTION 2

Some plants store starch as grains.

Some starch grains have rings on the surface and the grains from different plant types differ in size and shape.

You are required to:

- observe and draw starch grains from two different types of plant
- compare the starch grains from these two different types of plant.

You are provided with starch grains, in water, from two different types of plant, labelled **F** and **G**.

1. Put one **clean and dry** microscope slide on a piece of black card with a paper towel underneath as shown in Fig. 2.1.

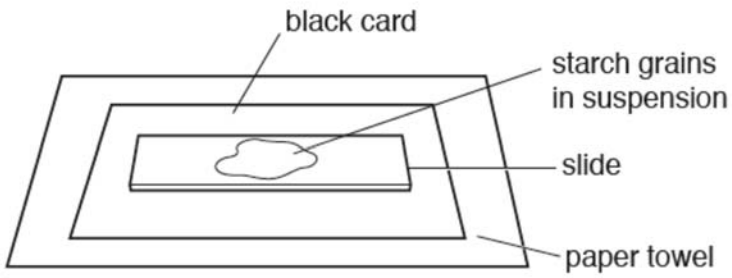

Fig. 2.1

2. Using a pipette, put a few drops of **F** containing starch grains onto the slide.

3. Cover the starch grains with a coverslip and use a paper towel to remove any excess liquid that is outside the coverslip.

4. View the slide to observe the starch grains using the microscope. Select an area of starch grains on the slide.

You may need to reduce the amount of light entering the microscope to observe the grains clearly.

You must adjust the fine focus to observe the starch grains.

5. Select **three** starch grains which show the different sizes and features of the grains.

6. Make a large drawing in **(a)(i)** of the three starch grains that you have selected.

7. Remove the slide from the microscope and place on a paper towel.

8. Repeat step 1 to step 7 with **G**.

Use a sharp pencil for drawing.

(a) (i) Make a large drawing of the three starch grains from **F** and the three starch grains from **G**.

Use **one** ruled label line and label **X** to identify the surface markings of **one** starch grain.

*starch grains from **F***

*starch grains from **G***

[4]

(ii) Annotate your drawings in **(a)(i)** to describe **three** observable differences between the starch grains from **F** and **G** by:

- drawing label lines to each of the features on the starch grains that show these differences
- describing next to each line how each feature is different.

[3]

The eyepiece graticule scale in your microscope may be used to measure the actual length of the layers of tissue or cells, if the scale has been calibrated against a stage micrometer.

You are provided with:

- Microscope with an eyepiece graticule fitted into the eyepiece lens
- Stage Micrometer: 1 mm divided into 100 divisions. Each is 0.01mm.

(b) (i) Using the appropriate objective lens and the eyepiece graticule, find the actual width, in µm, of the three starch grains that you have drawn from **F**

You may lose marks if you do not show your working.

Working for the calibration of the eyepiece graticule scale for the appropriate objective lens that you have selected:

Working for the measurement of the starch grains that you have drawn from **F**:

actual width of starch grains: 1………………… 2………………… 3…………………

[5]

(ii) Using the actual widths calculated in **(i)**, calculate the **mean** actual width of a starch grain that you have drawn from **F**.

You may lose marks if you do not show your working or if you do not use appropriate units.

mean actual width ………………………………[2]

Fig. 2.2 is a photomicrograph of a stained transverse section through part of a leaf of the plant from which starch grains labelled **G** were taken from.

You are not expected to be familiar with this specimen.

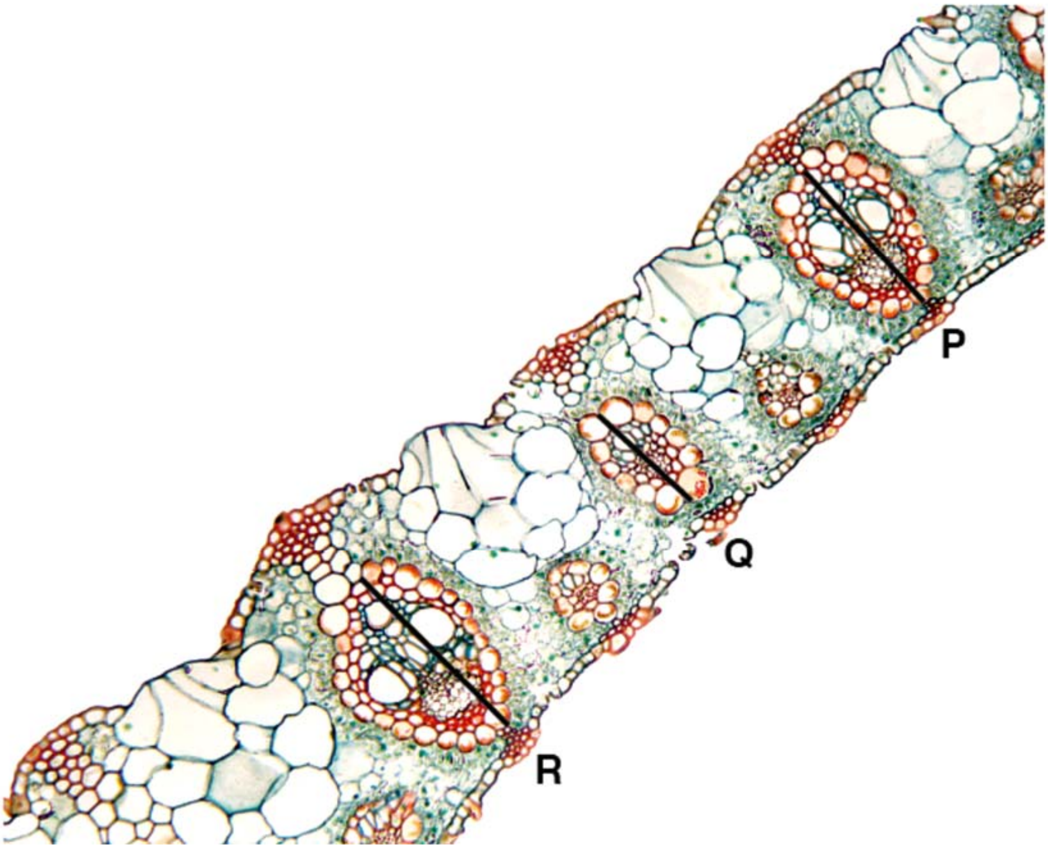

Fig. 2.2

Observe the different tissues in Fig. 2.2 and select an area that shows part of the epidermis and the vascular bundles.

Use a sharp pencil for drawing.

(c) Draw **one** large plan diagram from the selected area which shows:

- part of the epidermis
- only **three** larger vascular bundles labelled **R**, **Q** and **P**
- only **two** smaller vascular bundles
- any other observable tissues.

Use **one** ruled label line and label to identify the phloem.

[4]

[Total: 18]

Note:

For those candidates who started with Question 2, start Question 1 as soon as you have completed Question 2.

QUESTION 3

The information on the label of a bottle of one brand of cleaning solution for contact lenses states that it contains enzymes which remove protein from contact lenses.

The instructions for using the solution state that:

- The cleaning solution should be poured into the cleaning pot provided up to the marked level.
- The contact lenses should be left in the cleaning solution for a minimum of four hours.
- Before using the lenses, they should be washed in a sterile saline (sodium chloride) solution.

The contents of the solution are:

1. Subtilisin A, a protease with an optimum temperature of 60 °C and an optimum pH of 7.0–7.5.
2. Buffered saline, a physiological solution with the same ‘balance of ions’ as body fluids.
3. Disodium EDTA, a preservative.

A student was asked to find out the actual concentration of the protease present in the contact lens cleaning solution.

The student found a web site that gave the concentration of subtilisin A in different brands of contact lens cleaning solutions. These range between 20 μg cm^{-3} and 100 μg cm^{-3}.

The student simulated a dirty contact lens using the protein gelatin and a thin transparent plastic sheet. One side of the plastic sheet was dipped into melted gelatin containing a dye. The gelatin was then allowed to set.

Fig. 3.1 shows a simulated contact lens.

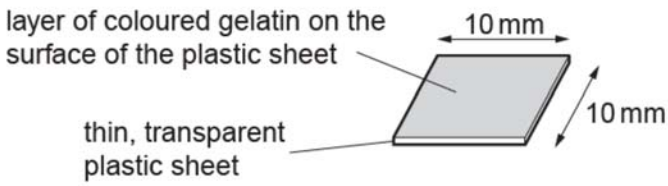

Fig. 3.1

The student tested the activity of subtilisin A by recording the time taken for the coloured gelatin to be removed, leaving the transparent plastic sheet.

Fig. 3.2 shows the apparatus the student used.

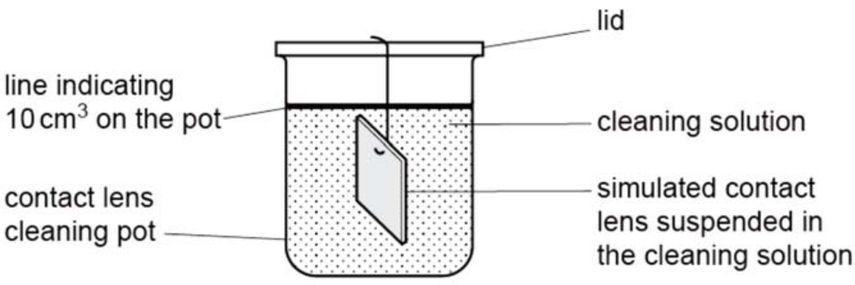

Fig. 3.2

Table 3.1 shows the results of trials to find the time taken for the coloured gelatin to be removed by the cleaning solution at 35 °C.

Table 3.1

trial	1	2	3
time / minutes	35	50	40

(a) The student noticed that there was a large variation in the results. This may be because it was difficult to know when to stop the stopwatch.

Suggest two problems in deciding when to stop the stopwatch in this investigation.

....

....

....

.... [2]

(b) The student made a stock cleaning solution containing 1 mg cm^{-3} of subtilisin A and a buffered saline solution containing disodium EDTA. The buffered saline solution containing disodium EDTA was the same concentration as used in the trials.

This solution was used to make a range of concentrations that could be used to find the concentration of subtilisin A in the contact lens cleaning solution.

The student decided to make 50 cm^3 of each concentration.

The student used the apparatus shown in Fig. 3.2 to investigate the effect of different concentrations of subtilisin A on the rate of removal of coloured gelatin from the simulated lenses at 35 °C.

Using all the above information given and your own knowledge, design an experiment that the student could use to find the following:

(i) the effect of different concentrations of subtilisin A on the removal of protein from the simulated lenses and

(ii) the actual concentration of subtilisin A in the contact lens cleaning solution.

You must use:

- stock cleaning solution containing 1 mg cm^{-3} of subtilisin A and a buffered saline solution containing disodium EDTA. The buffered saline solution containing disodium EDTA was the same concentration as used in the trials.
- buffered saline solution containing disodium EDTA
- plastic sheets (10 mm x 10 mm) previously dipped into melted gelatin containing a dye and the gelatin has been allowed to set.
- distilled water
- thermostatically controlled water bath
- stopwatch

You may select from the following apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rod, etc.
- syringes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[14]

[Total: 16]